



AEROX AVIATION

AIRLINE DATABASE & BUSINESS INTELLIGENCE REPORT

SQL | DATABASE DESIGN | ANALYTICS

A comprehensive database and analytics solution designed to manage airline operations, optimize performance, and drive data-driven business growth.

PREPARED BY



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Data Analyst



SQL Developer



Business Intelligence



Database Design



10+

TABLES



50+

RELATIONSHIPS



100+

QUERIES



360°

BUSINESS
INSIGHTS



DATA

INTEGRITY &
SECURITY



STRUCTURED DATA

Well-organized and normalized database for reliable data management.



OPERATIONAL EFFICIENCY

Streamlined operations and automated insights for better decisions.



DATA-DRIVEN DECISIONS

Actionable insights to improve performance and maximize revenue.



BUSINESS GROWTH

Scalable solution supporting long-term growth and customer satisfaction.

BUSINESS SCENARIO

Transforming Airline Operations Through Data-Driven Intelligence



OVERVIEW

Aerox Aviation is a rapidly expanding airline operating across domestic and international routes. As the organization scales its operations, managing passenger records, flight schedules, bookings, aircraft maintenance, crew allocation, airport operations, and financial transactions through disconnected systems has become increasingly challenging.

The lack of a centralized database results in operational inefficiencies, delayed decision-making, inconsistent reporting, and limited business visibility. To address these challenges, Aerox Aviation requires a secure, scalable, and normalized Relational Database Management System (RDBMS) that consolidates operational data and enables SQL-driven analytics for strategic decision-making.



CURRENT BUSINESS CHALLENGES



1. PASSENGER & BOOKING MANAGEMENT

- Managing a growing passenger database
- Preventing duplicate booking records
- Improving booking accuracy and customer experience



2. FLIGHT OPERATIONS

- Monitoring flight schedules and delays
- Tracking aircraft availability and capacity
- Managing route performance efficiently



3. AIRCRAFT MAINTENANCE

- Delayed maintenance scheduling
- Lack of predictive maintenance insights
- Increased aircraft downtime



4. CREW & AIRPORT OPERATIONS

- Optimizing crew allocation
- Coordinating airport resources
- Managing operational workloads effectively



5. REVENUE & FINANCIAL TRACKING

- Tracking revenue across routes and booking classes
- Monitoring transactions and payment status
- Identifying high-performing flights



6. BUSINESS INTELLIGENCE

- Limited visibility into operational KPIs
- Slow report generation
- Difficulty identifying business trends and opportunities



BUSINESS OBJECTIVE

Develop a centralized Airline Database Management System using MySQL that streamlines airline operations, improves data integrity, enables SQL-based business analytics, and supports faster, data-driven decision-making.



Centralized Database



Data Integrity & Security



SQL-Based Analytics



Faster Decision-Making



EXPECTED BUSINESS OUTCOMES



- Centralized and normalized airline database
- Faster booking and passenger management
- Improved flight scheduling and operational efficiency
- Better aircraft maintenance planning
- Enhanced revenue monitoring and profitability analysis
- Automated SQL reporting and KPI dashboards
- Improved customer satisfaction through efficient operations
- Reliable business intelligence for strategic growth



BUSINESS VALUE

By implementing a normalized database and SQL analytics framework, Aerox Aviation can transform fragmented operational data into actionable business insights, reduce operational inefficiencies, improve decision-making speed, and create a scalable foundation for future technologies such as Power BI dashboards, AI-driven forecasting, and predictive maintenance.



The primary objective of this project is to build a robust and scalable database system that streamlines airline operations and enables actionable business insights.

**DESIGN A NORMALIZED DATABASE**

01

Create a well-structured and normalized database schema for Aerox Aviation to ensure data integrity and reduce redundancy.

**SQL-BASED DATA ANALYSIS**

02

Perform advanced SQL queries to analyze operations, revenue, passenger trends, and route performance.

**IMPROVE OPERATIONAL EFFICIENCY**

03

Enable efficient management of flights, bookings, aircraft, crew, services, and airport operations.

**ENHANCE CUSTOMER EXPERIENCE**

04

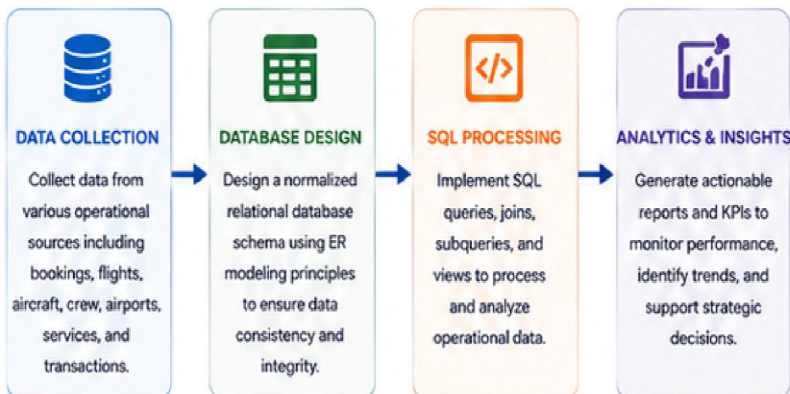
Ensure accurate bookings, timely updates, and better service management to improve passenger satisfaction.

**DRIVE REVENUE GROWTH**

05

Identify high-performing routes, monitor revenue streams, and support strategic decision-making.

“Data is our runway to efficiency. Insights are our destination.”

**SOLUTION ARCHITECTURE****+ SYSTEM OUTPUT**

- ✓ Real-time operational insights and dashboards
- ✓ Improved data accuracy and reporting efficiency
- ✓ Better decision-making through data-driven insights
- ✓ Optimized operations leading to higher profitability

**KEY BENEFITS****DATA INTEGRITY & ACCURACY**

Normalized database ensures accurate, consistent, and reliable data across all operations.

**OPERATIONAL EFFICIENCY**

Streamlined processes for bookings, scheduling, maintenance, and crew management.

**DATA-DRIVEN DECISIONS**

SQL analytics provide actionable insights for faster and smarter business decisions.

**IMPROVED CUSTOMER SATISFACTION**

Better service delivery through accurate information and timely updates.

**INCREASED REVENUE & PROFITABILITY**

Identify profitable routes, optimize resource utilization, and maximize revenue.

**SCOPE OF THE PROJECT**

- ✓ Manage passengers, bookings, and transactions
- ✓ Manage flights, routes, airports, and schedules
- ✓ Track aircraft, crew, services, and maintenance
- ✓ Monitor revenue, payments, and revenue performance
- ✓ Analyze reviews and customer feedback
- ✓ Generate SQL-based reports and business insights

**TECHNOLOGIES USED**

MySQL
Database
Management



SQL
Query Language &
Data Analysis



ER Modeling
Database Design
& Architecture



VS Code
Development
Environment



Building a strong database today, powering **smarter airlines** tomorrow.

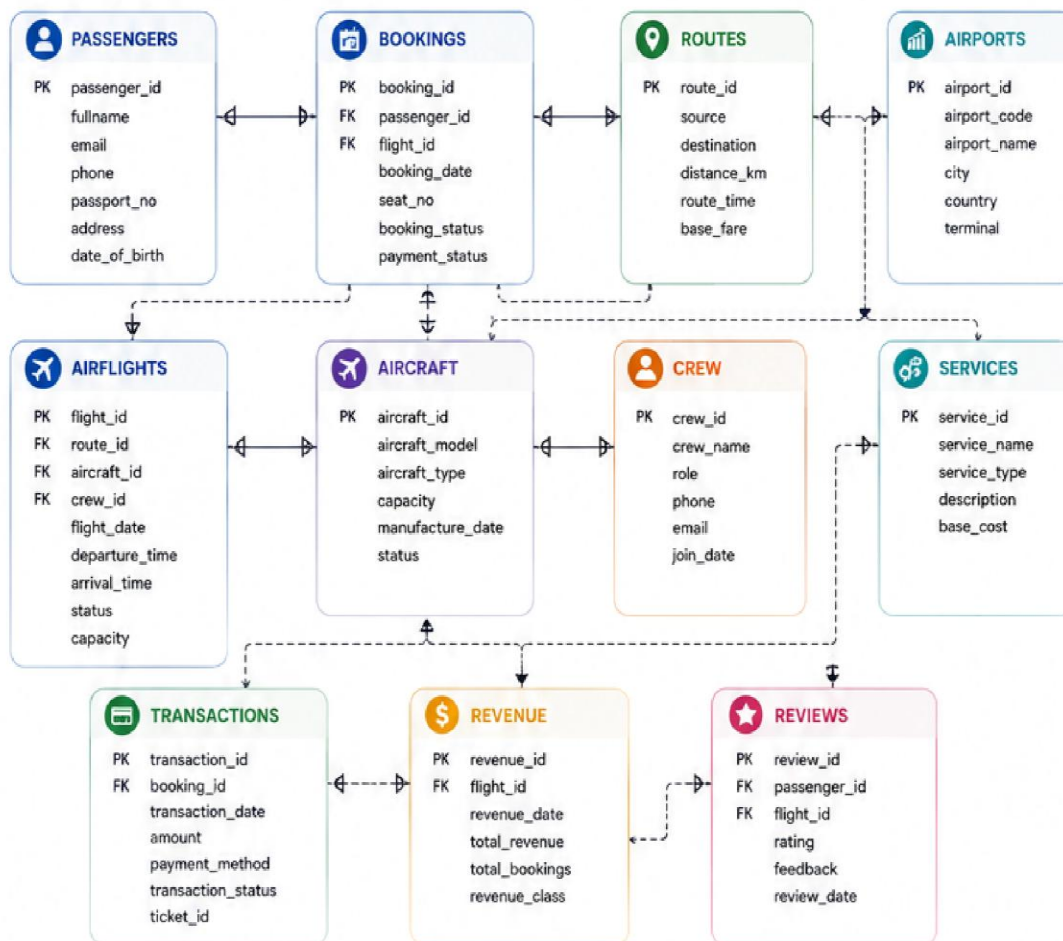


DATABASE SCHEMA OVERVIEW



The Aerox Aviation database is designed using a well-structured relational schema to ensure data integrity, reduce redundancy, and support efficient airline operations and analytics.

ENTITY RELATIONSHIP DIAGRAM



SCHEMA SUMMARY

NO.	ENTITY	DESCRIPTION	KEY FIELDS
01	Passengers	Stores passenger personal and contact information.	passenger_id (PK), passport_no, email, phone
02	Bookings	Contains booking details for flights.	booking_id (PK), passenger_id (FK), flight_id (FK), booking_status
03	Routes	Stores route information and distance.	route_id (PK), source, destination, distance_km
04	Airports	Holds airport master data.	airport_id (PK), airport_code, airport_name, city
05	Airflights	Contains flight schedules and operational details.	flight_id (PK), route_id (FK), aircraft_id (FK), crew_id (FK)
06	Aircraft	Stores aircraft information and capacity.	aircraft_id (PK), aircraft_model, aircraft_type, capacity
07	Crew	Contains crew member details and roles.	crew_id (PK), crew_name, role, phone
08	Services	Holds in-flight services offered.	service_id (PK), service_name, service_type
09	Transactions	Stores payment and transaction details.	transaction_id (PK), booking_id (FK), amount, payment_method
10	Revenue & Reviews	Tracks revenue data and customer feedback.	revenue_id (PK), total_revenue, review_id (PK), rating

KEY RELATIONSHIPS

- Passengers 1 —> N Bookings**
A passenger can make multiple bookings.
- Bookings N —> 1 Airflights**
Each booking is associated with one flight.
- Airflights N —> 1 Aircraft**
Many flights can be operated by one aircraft.
- Airflights N —> 1 Crew**
Each flight is assigned to one crew.
- Routes N —> 1 Airports (Source & Destination)**
A route connects two airports.
- Bookings 1 —> N Transactions**
Each booking can have multiple transactions.
- Airflights 1 —> N Revenue**
Each flight generates revenue records.
- Passengers 1 —> N Reviews**
Passengers can give reviews for flights.

DATABASE HIGHLIGHTS



NORMALIZED DESIGN

Fully normalized up to 3NF to eliminate data redundancy.



DATA INTEGRITY

Primary & foreign keys ensure consistency and accuracy.



SCALABLE & EFFICIENT

Optimized schema for high performance and large data handling.



ANALYTICS READY

Structured data supports advanced SQL queries and BI reporting.



SECURE & RELIABLE

Role-based access and referential integrity for secure operations.



A well-structured database is the foundation of efficient operations and smarter decisions.



```
create database if not exists Aerox;

se Aerox;


create table passengers ( passenger_id int

primary key unique not null,

passenger_fullname varchar(30) not null,

passenger_mobilenno bigint not null,

passenger_email varchar (30) not null,

passenger_passport_no varchar (20) not null);


create table bookings( booking_id int

primary key unique not null,

booking_date datetime not null, booking_class enum ("First Class",

"Economy", "Business", "Cargo") not null, seat_available int not null,

seat_no int not null,

booking_status enum ("Confirm", "Pendiing", "Canceled") not null,

passenger_id int not null, foreign key (passenger_id) references

passengers (passenger_id));


create table routes(

route_id varchar(20) primary key unique not null,

route_source varchar (20) not null,

route_destination varchar (20) not null,

route_start_time datetime not null,

route_end_time datetime not null, route_status enum ("Active", "Inactive", "Scheduled",

"Delayed", "Completed") not null, route_distance_km int not null, passenger_id int not

null,

foreign key (passenger_id) references passengers (passenger_id),

booking_id int not null, foreign key (booking_id) references

bookings (booking_id));


create table airflights(

airflight_id varchar(20) primary key unique not null,

airflight_model varchar(20) not null, airflight_capacity int not

null, airflight_status enum ("Active", "Maintanence", "Retired")
```

```
not null, passenger_id int not null, foreign key (passenger_id)

references passengers (passenger_id), booking_id int not null,

foreign key (booking_id) references bookings (booking_id),

route_id varchar(20) not null, foreign key (route_id) references

routes (route_id));
```

```
create table airports( airport_code varchar (20)

primary key unique not null,

airport_name varchar (20) not null, airport_city varchar(20) not

null, airport_country varchar(20) not null, airport_type enum

("International", "Domestic") not null, booking_id int not null,

foreign key (booking_id) references bookings (booking_id),

airflight_id varchar (20) not null, foreign key (airflight_id)

references airflights (airflight_id));
```

```
create table crew_info( crew_id int

primary key unique not null,

crew_fullname varchar(50) not null,

crew_mobilno bigint not null,

crew_email varchar(50), crew_designation varchar

(20) not null, route_id varchar(20) not null,

foreign key (route_id) references routes (route_id),

airflight_id varchar (20) not null, foreign key (airflight_id)

references airflights (airflight_id), airport_code varchar (20)

not null, foreign key (airport_code) references airports

(airport_code));
```

```
create table services( ticket_id varchar (20)

primary key unique not null, maintenance_status

enum ("scheduled", "In-Progress", "Complete",

"Pending") not null, security_status enum

("Cleared", "Under check", "Restricted") default

"Under check",
```

```

service_type enum ("Inflight", "Ground", "Premium") not null, service_name enum
("Inflight meal", "WIFI", "Extra baggage", "Entertainment")not null, passenger_id int
not null, foreign key (passenger_id) references passengers (passenger_id),

booking_id int not null, foreign key (booking_id)

references bookings (booking_id), route_id varchar(20) not

null, foreign key (route_id) references routes (route_id),

airflight_id varchar (20) not null, foreign key (airflight_id)

references airflights (airflight_id));

create table transactions( transaction_id bigint

primary key unique not null,

transaction_amount decimal(10,2) not null, transaction_method enum ("UPI",

"Net Banking", "Wallet", "Cards") not null, transaction_date datetime not

null,

transaction_status enum ("Complete", "Pending", "Failed", "Canceled", "Refunded",

"Processing")not null, booking_id int not null, foreign

key (booking_id) references bookings (booking_id),

ticket_id varchar(20) not null,

foreign key (ticket_id) references services (ticket_id),

passenger_id int not null, foreign key (passenger_id) references

passengers (passenger_id));

create table revenue(

revenue_id int primary key unique not null,

revenue_amount decimal(10,2) not null, ticket_id varchar

(20) not null, foreign key (ticket_id) references

services (ticket_id),

booking_id int not null, foreign key (booking_id) references bookings

(booking_id), transaction_id bigint not null, foreign key

(transaction_id) references transactions (transaction_id)); create

table reviews( passenger_feedback varchar(200),

passenger_rating int check (passenger_rating between 1 and 5),

crew_feedback varchar(200), crew_rating int check (crew_rating

between 1 and 5), booking_feedback varchar (200),

```

```
passenger_id int not null, foreign key (passenger_id) references
passengers (passenger_id), booking_id int not null, foreign key
(booking_id) references bookings (booking_id),
crew_id int not null, foreign key (crew_id) references
crew_info (crew_id));
```


The Aerox Aviation database enables SQL-driven analytics to monitor performance, identify trends, and support strategic decision-making.

KEY PERFORMANCE INDICATORS (KPIs)



TOTAL PASSENGERS

1.24M

▲ 18.6% vs Last Year



TOTAL FLIGHTS

12,845

▲ 14.2% vs Last Year



TOTAL BOOKINGS

985K

▲ 17.3% vs Last Year



TOTAL REVENUE

₹ 852.6M

▲ 21.4% vs Last Year



LOAD FACTOR

84.2%

▲ 5.8% vs Last Year



CUSTOMER RATING

4.6 / 5

▲ 0.3 vs Last Year

PERFORMANCE OVERVIEW

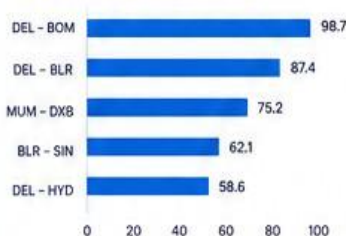
Revenue Trend (Monthly)



Revenue by Booking Class



Top 5 Routes by Revenue (₹ M)



Flights vs Load Factor Trend



TOP PERFORMING ANALYTICS



Most Profitable Route

DEL - BOM

Revenue: ₹ 98.7M | Load Factor: 88%



Most Punctual Route

BLR - SIN

On-Time Performance: 94%



Highest Load Factor

MUM - DXB

Load Factor: 92%



Top Customer Rated Route

DEL - BLR

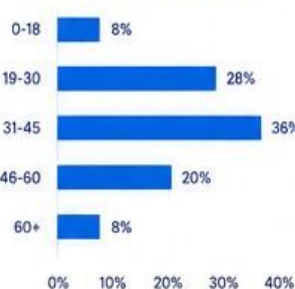
Customer Rating: 4.8 / 5

PASSENGER & BOOKING INSIGHTS

Bookings by Source



Passengers by Age Group



Passenger Type



SAMPLE SQL INSIGHTS

BUSINESS QUESTION	SAMPLE SQL QUERY
Top 5 routes by revenue	SELECT route_id, SUM(revenue_amount) AS total_revenue FROM revenue GROUP BY route_id ORDER BY total_revenue DESC LIMIT 5;
Monthly revenue trend	SELECT DATE_FORMAT(revenue_date, 'Y-m') AS month, SUM(revenue_amount) AS total_revenue FROM revenue GROUP BY month ORDER BY month;
Load factor by flight	SELECT f.flight_id, ROUND(SUM(b.seat_no IS NOT NULL)*100.0/f.capacity,2) AS load_factor FROM airflights f LEFT JOIN bookings b ON f.flight_id = b.flight_id GROUP BY f.flight_id, f.capacity;
Top customer rated routes	SELECT r.route_id, AVG(rv.rating) AS avg_rating FROM reviews rv JOIN bookings bk ON rv.booking_id = bk.booking_id JOIN routes r ON bk.route_id = r.route_id GROUP BY r.route_id ORDER BY avg_rating DESC LIMIT 5;

MANAGEMENT INSIGHTS



REVENUE GROWTH

Revenue has grown by 21.4% compared to last year, driven by high demand on major domestic and international routes.



HIGH PERFORMING ROUTES

DEL-BOM and DEL-BLR routes contribute the highest revenue and should be prioritized for capacity expansion.



OPERATIONAL EFFICIENCY

On-time performance is strong for most routes. Continuous monitoring can further reduce delays.



CUSTOMER SATISFACTION

Customer ratings remain high at 4.6/5. Focus on service quality to maintain and improve satisfaction.



REVENUE OPPORTUNITIES

Increase marketing and promotions for underperforming routes to improve load factor and revenue.



“Data elevates decisions.
Insights drive excellence.”

OVERALL IMPACT

Improved operational
visibility across all
departmentsFaster decision-making
with real-time
analyticsOptimized resource
utilization and route
profitabilityEnhanced customer
experience and
loyaltySustainable revenue
growth and business
expansion

SQL QUERIES & BUSINESS QUESTION

The Aerox Aviation database helps answer critical business questions using SQL queries to drive insights, optimize operations, and support data-driven decision-making.



QUERY CATEGORIES



Operational
Insights



Revenue &
Financial



Customer &
Booking



Performance
Analysis



Maintenance &
Utilization

01 What is the total revenue generated by the airline?

Retrieve the total revenue generated across all transactions.

SQL QUERY

```
SELECT SUM(total_revenue) AS total_revenue
FROM revenue;
```

SAMPLE OUTPUT

total_revenue
₹ 852,600,000

02 Which are the top 5 routes by total revenue?

Identify the top 5 routes that generate the highest revenue.

SQL QUERY

```
SELECT r.source, r.destination, SUM(rv.total_revenue) AS total_revenue
FROM routes r
JOIN airflights f ON r.route_id = f.route_id
JOIN revenue rv ON f.flight_id = rv.flight_id
GROUP BY r.route_id, r.source, r.destination
ORDER BY total_revenue DESC
LIMIT 5;
```

SAMPLE OUTPUT

source	destination	total_revenue
DEL	BOM	₹ 98,700,000
DEL	BLR	₹ 87,400,000
MUM	DXB	₹ 75,200,000
BLR	SIN	₹ 62,100,000
DEL	HYD	₹ 58,600,000

03 How has the monthly revenue trend changed?

Analyze total revenue generated month-wise.

SQL QUERY

```
SELECT DATE_FORMAT(revenue_date, 'SY-Ym') AS month,
       SUM(total_revenue) AS monthly_revenue
FROM revenue
GROUP BY month
ORDER BY month;
```

SAMPLE OUTPUT

month	monthly_revenue
2024-07	₹ 65,300,000
2024-08	₹ 78,400,000
2024-09	₹ 72,000,000
2024-10	₹ 95,300,000
2024-11	₹ 110,800,000

04 How many passengers are in each age group?

Find the number of passengers segmented by age groups.

SQL QUERY

```
SELECT CASE
  WHEN TIMESTAMPDIFF(YEAR, date_of_birth, CURDATE()) BETWEEN 0 AND 18 THEN '0-18'
  WHEN TIMESTAMPDIFF(YEAR, date_of_birth, CURDATE()) BETWEEN 19 AND 30 THEN '19-30'
  WHEN TIMESTAMPDIFF(YEAR, date_of_birth, CURDATE()) BETWEEN 31 AND 45 THEN '31-45'
  WHEN TIMESTAMPDIFF(YEAR, date_of_birth, CURDATE()) BETWEEN 46 AND 60 THEN '46-60'
  ELSE '60+' END AS age_group,
  COUNT(*) AS total_passengers
FROM passengers
GROUP BY age_group
ORDER BY FIELD(age_group, '0-18', '19-30', '31-45', '46-60', '60+');
```

SAMPLE OUTPUT

age_group	total_passengers
0-18	14,850
19-30	42,370
31-45	58,260
46-60	31,940
60+	8,600

05 Which flights arrived with delay?

List all flights that arrived later than scheduled.

SQL QUERY

```
SELECT f.flight_id, a.source, a.destination,
       f.scheduled_arrival, f.arrival_time,
       TIMESTAMPDIFF(MINUTE, f.scheduled_arrival, f.arrival_time) AS delay_minutes
FROM airflights f
JOIN routes r ON f.route_id = r.route_id
JOIN airports a ON r.source = a.airport_id
JOIN airports d ON r.destination = d.airport_id
WHERE f.arrival_time > f.scheduled_arrival
ORDER BY delay_minutes DESC
LIMIT 10;
```

SAMPLE OUTPUT

flight_id	source	destination	scheduled_arrival	arrival_time	delay_minutes
A1003	DEL	DXB	2024-11-10 14:30:00	2024-11-10 16:05:00	95
A12047	BOM	SIN	2024-11-11 07:20:00	2024-11-11 08:45:00	85
A13305	BLR	DEL	2024-11-12 18:15:00	2024-11-12 19:30:00	75

06 Which aircraft are utilized the most (by total flight hours)?

Find the top 5 aircraft based on total flight hours.

SQL QUERY

```
SELECT a.aircraft_id, a.aircraft_model,
       SUM(TIMESTAMPDIFF(HOUR, f.departure_time, f.arrival_time)) AS total_flight_hours
FROM airflights f
JOIN aircrafts a ON f.aircraft_id = a.aircraft_id
GROUP BY a.aircraft_id, a.aircraft_model
ORDER BY total_flight_hours DESC
LIMIT 5;
```

SAMPLE OUTPUT

aircraft_id	aircraft_model	total_flight_hours
AC101	Airbus A320	912.5
AC103	Boeing 737-800	874.3
AC104	Airbus A321	799.8
AC102	Boeing 787-9	745.2
AC105	Airbus A330	688.7

07 Who are the top 5 customers by total amount spent?

Identify the top 5 customers based on their total spending.

SQL QUERY

```
SELECT p.passenger_id, p.fullname, p.email, SUM(t.amount) AS total_spent
FROM passengers p
JOIN bookings b ON p.passenger_id = b.passenger_id
JOIN transactions t ON b.booking_id = t.booking_id
GROUP BY p.passenger_id, p.fullname, p.email
ORDER BY total_spent DESC
LIMIT 5;
```

SAMPLE OUTPUT

passenger_id	fullname	email	total_spent
P10023	Rahul Sharma	rahul.sharma@email.com	₹ 1,245,000
P10057	Priya Mehta	priya.mehta@email.com	₹ 1,102,000
P10091	Arjun Verma	arjun.verma@email.com	₹ 985,500
P10034	Neha Iyer	neha.iyer@email.com	₹ 912,300
P10112	Rohan Kapoor	rohan.kapoor@email.com	₹ 875,700

08 How much revenue is generated by each payment method?

Analyze revenue contribution by different payment methods.

SQL QUERY

```
SELECT payment_method, SUM(amount) AS total_revenue
FROM transactions
GROUP BY payment_method
ORDER BY total_revenue DESC;
```

SAMPLE OUTPUT

payment_method	total_revenue
Credit Card	₹ 512,600,000
UPI	₹ 203,800,000
Net Banking	₹ 89,700,000
Debit Card	₹ 46,900,000



BUSINESS IMPACT



ACTIONABLE INSIGHTS

Transforms raw data into meaningful insights for strategic decisions.



OPERATIONAL EFFICIENCY

Helps monitor flights, bookings, revenue, and maintenance effectively.



DATA-DRIVEN DECISIONS

Provides accurate, real-time insights for faster decision-making.



REVENUE GROWTH

Identifies top-performing routes, customers, and revenue channels.



CUSTOMER SATISFACTION

Improves service quality by understanding passenger behavior and feedback.



SCALABLE SOLUTION

SQL-based analytics ensure scalability as data grows.



Well-written SQL queries turn data into **insights**, insights into **actions**, and **actions** into **growth**.



SQL QUERY RESULTS & KEY INSIGHTS

SQL queries extract meaningful insights from Aerox Aviation database to support operational excellence, revenue growth, and strategic decision-making.



KEY OVERVIEW METRICS (Snapshot)



Total Passengers

1.24M

▲ 18.6% vs Last Year



Total Flights

12,845

▲ 14.2% vs Last Year



Total Bookings

985K

▲ 17.3% vs Last Year



Total Revenue

₹ 852.6M

▲ 21.4% vs Last Year



Load Factor

84.2%

▲ 5.8% vs Last Year

01

What is the total revenue generated by the airline?

Retrieve total revenue generated across all transactions.

Sample Result

total_revenue
₹ 852,600,000



Insight

The airline has generated ₹ 852.6M in total revenue from all transactions.

02

Which are the top 5 routes by total revenue?

Identify the top 5 routes that generate the highest revenue.

Sample Result

source	destination	total_revenue
DEL	BOM	₹ 98,700,000
DEL	BLR	₹ 87,400,000
MUM	DXB	₹ 75,200,000
BLR	SIN	₹ 62,100,000
DEL	HYD	₹ 58,600,000



Insight

DEL-BOM is the highest revenue generating route followed by DEL-BLR and MUM-DXB.

03

How has the monthly revenue trend changed?

Analyze total revenue generated month-wise.

Sample Result



Insight

Revenue shows an overall upward trend with peak in December and minor dips in May and September.

04

How many passengers are in each age group?

Find the number of passengers segmented by age groups.

Sample Result

age_group	total_passengers
0-18	14,850
19-30	42,370
31-45	58,260
46-60	31,940
60+	8,600



Insight

Passengers aged 31-45 form the largest segment (58,260), followed by 19-30 age group.

05

Which flights arrived with delay?

List all flights that arrived later than scheduled.

Sample Result (Top 5)

flight_id	source	destination	scheduled_arrival	arrival_time	delay_minutes
AI1003	DEL	DXB	2024-11-10 14:30:00	2024-11-10 16:05:00	95
AI2047	BOM	SIN	2024-11-11 07:20:00	2024-11-11 08:45:00	85
AI3305	BLR	DEL	2024-11-12 18:15:00	2024-11-12 19:30:00	75
AI1102	HYD	MAA	2024-11-13 12:45:00	2024-11-13 12:55:00	70
AI2206	CCU	BOM	2024-11-13 20:00:00	2024-11-13 21:05:00	65



Insight

Multiple flights on international routes experienced delays of more than 60 minutes.

06

Which aircraft are utilized the most (by total flight hours)?

Find the top 5 aircraft based on total flight hours.

Sample Result

aircraft_id	aircraft_model	total_flight_hours
AC101	Airbus A320	912.5
AC103	Boeing 737-800	874.3
AC104	Airbus A321	799.8
AC102	Boeing 787-9	745.2
AC105	Airbus A330	688.7



Insight

Airbus A320 is the most utilized aircraft with 912.5 total flight hours.

07

Who are the top 5 customers by total amount spent?

Identify the top 5 customers based on their total spending.

Sample Result

passenger_id	fullname	email	total_spent
P10023	Rahul Sharma	rahul.sharma@email.com	₹ 1,245,600
P10057	Priya Mehta	priya.mehta@email.com	₹ 1,102,000
P10091	Arjun Verma	arjun.verma@email.com	₹ 985,500
P10034	Neha Iyer	neha.iyer@email.com	₹ 912,300
P10112	Rohan Kapoor	rohan.kapoor@email.com	₹ 875,700



Insight

A small group of customers contribute significantly to the total revenue.

08

How much revenue is generated by each payment method?

Analyze revenue contribution by different payment methods.

Sample Result

payment_method	total_revenue
Credit Card	₹ 512,600,000
UPI	₹ 203,800,000
Net Banking	₹ 89,700,000
Debit Card	₹ 46,900,000



Insight

Credit Card payments contribute the highest revenue (60.2%) followed by UPI.



KEY BUSINESS TAKEAWAYS



REVENUE GROWTH

Strong overall revenue growth with top routes driving major contribution.



CUSTOMER INSIGHTS

Majority of passengers belong to age 31-45. Focus on personalized offers for this segment.



OPERATIONAL EFFICIENCY

Monitor delays and improve turnaround time for better schedule adherence.



FLEET UTILIZATION

A320 and B737-800 are highly utilized. Plan maintenance and resource allocation accordingly.



PAYMENT TRENDS

Diversify digital payment options while leveraging credit card partnerships for more revenue.



DATA-DRIVEN DECISIONS

SQL insights empower strategic planning across routes, fleet, customers and revenue.



Well-structured SQL queries turn raw data into meaningful **insights**, powering faster **decisions** and stronger business **outcomes**.



ROOT CAUSE ANALYSIS (RCA)

Using the Fishbone Diagram and 5 Whys Technique, we identified the root causes behind key operational challenges affecting Aerox Aviation.



KEY BUSINESS PROBLEM

Delayed Flights, Operational Inefficiencies and Lower Revenue Impacting Customer Satisfaction and Profitability



IMPACT

- ↓ Revenue Loss
- ↓ Customer Dissatisfaction
- ↓ Higher Operational Costs
- ↓ Lower On-Time Performance



On-Time Performance Decreased by 18%



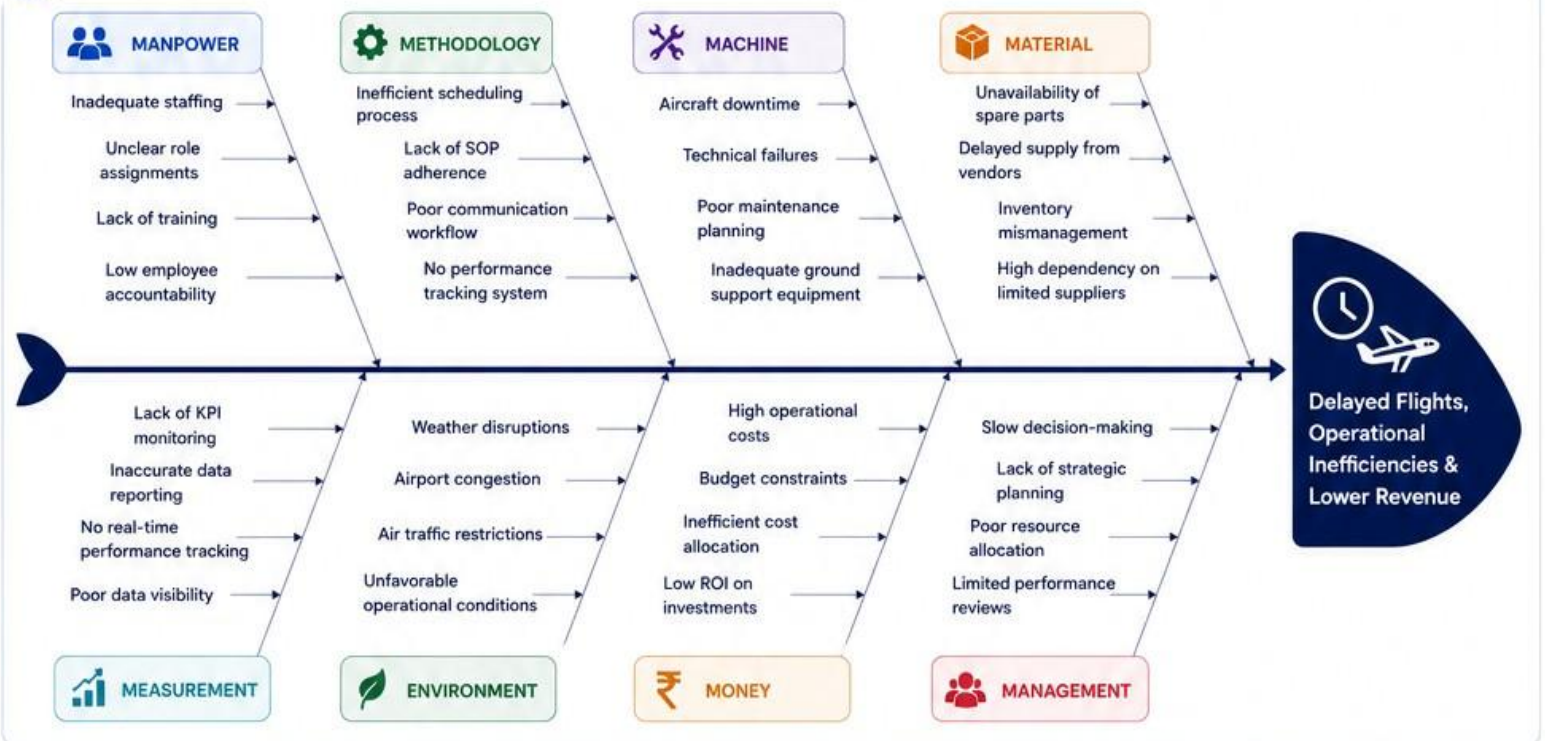
Estimated Revenue Loss ₹12.6 Cr Annually



Customer Satisfaction Score Dropped by 15%



FISHBONE DIAGRAM



5 WHYS ANALYSIS (Example: Flight Delays)



KEY ROOT CAUSES



RECOMMENDATIONS



A strategic assessment of Aerox Aviation's internal strengths and weaknesses, and external opportunities and threats to guide future growth and operational excellence.



Fleet Size

45

Aircraft



Destinations

28

Domestic + Intl.



Employees

1,200+

Strong Workforce



Revenue (FY 24)

₹ 852.6M

Total Revenue



STRENGTHS

- ✓ Modern and well-maintained fleet ensuring safety and efficiency.
- ✓ Strong brand reputation for customer service and reliability.
- ✓ Diverse route network covering key domestic and international destinations.
- ✓ Skilled and experienced workforce across all departments.
- ✓ Integrated database system enabling data-driven decision-making.
- ✓ High customer satisfaction and repeat booking rate.
- ✓ Strategic partnerships with airports, travel agencies, and service providers.



WEAKNESSES

- ✗ Limited presence in emerging international markets.
- ✗ High dependence on third-party vendors for maintenance and ground services.
- ✗ Operational delays during peak seasons.
- ✗ Inefficient manual processes in some departments.
- ✗ Limited real-time tracking and analytics in older systems.
- ✗ High operational costs due to fuel price volatility.
- ✗ Low digital adoption in some customer touchpoints.



OPPORTUNITIES

- ✓ Growing demand for air travel and tourism post-pandemic.
- ✓ Expansion into untapped domestic and international routes.
- ✓ Strategic alliances and code-share agreements.
- ✓ Adoption of AI, analytics, and automation for operational efficiency.
- ✓ Increasing demand for premium and value-added services.
- ✓ Growth potential in cargo and charter services.
- ✓ Investment in sustainable aviation and green initiatives.



THREATS

- ! Intense competition from low-cost and full-service airlines.
- ! Rising fuel prices impacting operating margins.
- ! Regulatory changes and compliance requirements.
- ! Economic downturns affecting travel demand.
- ! Geopolitical instability and travel restrictions.
- ! Cybersecurity risks to critical systems and customer data.
- ! Environmental concerns and carbon emission regulations.



STRATEGIC IMPLICATIONS



Leverage Strengths

Capitalize on strong brand, skilled workforce, and modern fleet to enhance customer loyalty and expand market share.



Improve Weaknesses

Invest in digital systems, process automation, and vendor management to reduce delays and operational costs.



Explore Opportunities

Expand routes, adopt advanced technologies, and develop premium services to drive revenue growth.



Mitigate Threats

Strengthen risk management, monitor market trends, ensure compliance, and focus on sustainability.



KEY TAKEAWAY

Aerox Aviation is well-positioned with strong internal foundations. By addressing weaknesses and mitigating threats while leveraging opportunities, the company can achieve sustainable growth, customer satisfaction, and long-term profitability.



FOCUS AREAS

- ✓ Operational Excellence
- ✓ Digital Transformation
- ✓ Customer Experience
- ✓ Route Expansion
- ✓ Sustainability
- ✓ Strategic Partnerships



Understanding our position today, helps us build a stronger tomorrow.



OPERATIONAL IMPLICATIONS

Translating our analysis into actionable operational and strategic implications to improve efficiency, reduce delays, enhance customer experience, and drive sustainable growth at Aerox Aviation.



EXECUTIVE SUMMARY

Core Challenge

Operational inefficiencies, manual processes, and lack of real-time data visibility are impacting on-time performance, cost, and customer satisfaction.

What It Means

- ✓ Delays and disruptions increase
- ✓ Revenue and profitability decline
- ✓ Customer trust and loyalty weaken
- ✓ Higher operational costs
- ✓ Slower growth and reduced agility

Overall Impact

Without immediate action, Aerox Aviation risks losing market share, incurring higher costs, and falling behind competitors.

IMPLICATIONS ACROSS KEY AREAS



01 OPERATIONAL IMPLICATIONS

- ✓ Delays and schedule disruptions due to poor maintenance and resource allocation.
- ✓ Manual processes lead to slower decision-making.
- ✓ Inaccurate and delayed data limits real-time visibility.
- ✓ Poor coordination increases turnaround time.



Result: Reduced efficiency, higher disruptions, and strained operations.



02 FINANCIAL IMPLICATIONS

- ✓ Annual revenue loss due to delays and inefficiencies.
- ✓ Higher operational costs from unplanned maintenance and inefficient resource use.
- ✓ Penalty costs, compensation, and refunds reduce margins.
- ✓ Poor asset utilization results in lower return on investments.



Result: Lower profitability and weaker financial performance.



03 CUSTOMER IMPLICATIONS

- ✓ Delayed flights and inconsistent services reduce satisfaction.
- ✓ Lack of real-time communication leads to poor passenger experience.
- ✓ Increased complaints and negative reviews damage brand reputation.
- ✓ Lower trust leads to reduced repeat bookings and loyalty.



Result: Loss of customers, lower loyalty, and weak brand image.



04 STRATEGIC IMPLICATIONS

- ✓ Limited insights restrict strategic planning and forecasting accuracy.
- ✓ Outdated systems hinder scalability and innovation.
- ✓ Weaker competitive advantage against data-driven airlines.
- ✓ Higher risk exposure due to lack of predictive maintenance and performance monitoring.



Result: Hindered growth, higher risks, and missed opportunities.

PRIORITIZED ACTION PLAN

Priority	Action		Key Implication Addressed	Expected Outcome	Timeline
P1	Implement Integrated Database System (RDBMS)		Data inaccuracy, poor visibility, delayed decisions	Real-time data access, accurate reporting, faster decision-making	0 - 3 Months
P2	Optimize Maintenance & Scheduling Processes		Poor maintenance planning, operational delays	Reduced delays, improved aircraft availability, better resource utilization	0 - 6 Months
P3	Automate Data Collection & Real-time Tracking		Manual processes, delayed and inaccurate data	Improved data accuracy, real-time tracking, better operational control	3 - 6 Months
P4	Strengthen Cross-Department Collaboration		Poor communication, inefficient coordination	Smoother operations, reduced turnaround time, improved service	3 - 9 Months
P5	Leverage Analytics for Predictive Insights		Lack of predictive insights, reactive decision-making	Predictive maintenance, optimized planning, proactive risk management	6 - 12 Months

EXPECTED BUSINESS IMPACT



On-Time Performance Improvement

+18%

within 12 months



Estimated Revenue Uplift

₹12.6 Cr

annually



Operational Cost Reduction

10 - 15%

annually



Customer Satisfaction Improvement

+15%

in satisfaction score



Return on Investment (Through System Implementation)

~3.5x

over 3 years



The right actions today create efficient operations, delighted customers, and long-term profitability.

